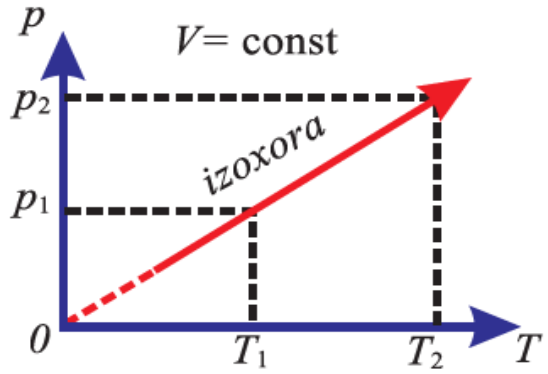
Fizika

9 sinf

MAVZU: Izoxorik jarayon



Ideal gazning massasi ($m=$ const) va hajmi $V = const$ o‘zgarmas bo‘lgandagi gaz holatining o‘zgarish jarayoniga ***izoxorik jarayon*** deyiladi.

Yunoncha “xoros” – hajm degan ma’noni anglatadi.

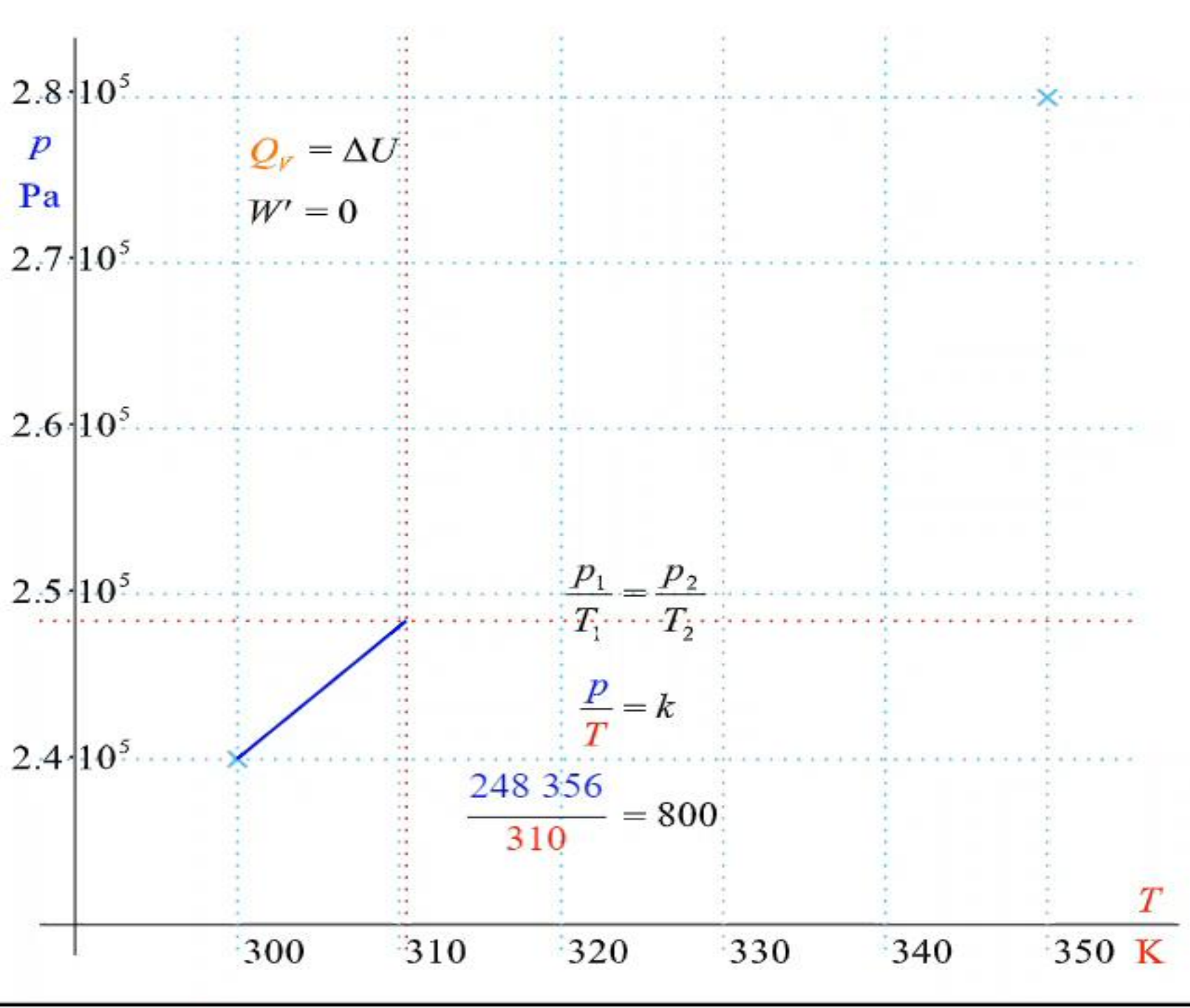
$$\begin{cases} P_1 \cdot V_1 = \frac{m}{M} \cdot R \cdot T_1 \\ P_2 \cdot V_2 = \frac{m}{M} \cdot R \cdot T_2 \end{cases} \Rightarrow V_1 = V_2 \text{ dan } \frac{P_1}{P_2} = \frac{T_1}{T_2}$$

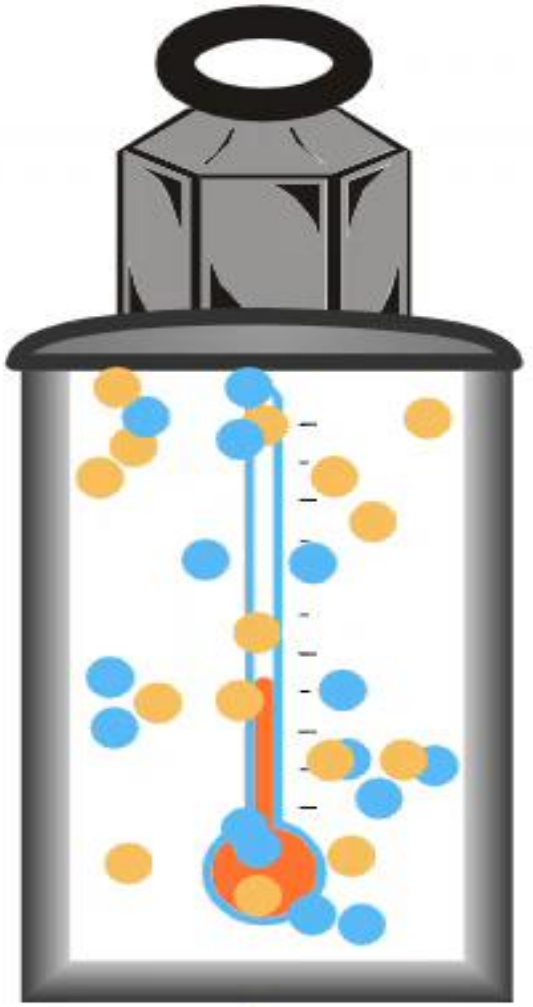
$$\frac{p}{T} = const$$

Demak, *izoxorik jarayonda berilgan massali gaz bosimining temperaturaga nisbati o‘zgarmas kattalik ekan.* Bu qonun 1787-yil fransuz fizigi *Jak Sharl* tomonidan tajribada topilganligi uchun ***Sharl qonuni*** deb ataladi.

$$\frac{p}{T} = const \Rightarrow p = const \cdot T$$

Ifodaga ko‘ra izoxorik jarayonda berilgan massali gaz bosimi uning absolyut tempera-turasiga to‘g‘ri proporsional ekan.


2.8 10^5
p
Pa
2.7 10^5
2.6 10^5
2.5 10^5
2.4 10^5
Q_V = ΔU
W' = 0
p_1/T_1 = p_2/T_2
p/T = k
248 356/310 = 800
T
300
310
320
330
340
350
K



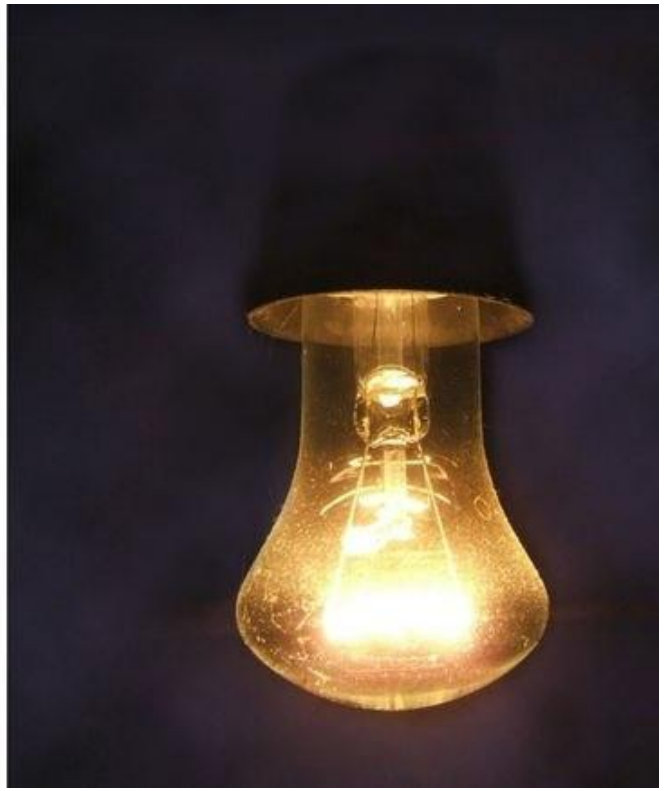




O‘zgarmas hajmda berilgan massali gaz bosimi temperaturaga to‘g‘ri proporsional ravishda o‘zgaradi.

Har qanday germetik yopiq idish yoki elektr lampochkasida isitilgan gaz bosimining ortishi *izoxorik* jarayon hisoblanadi.





Masala yechish namunasi

Gaz 280 K dan 540 K gacha izoxorik qizdirilganda uning bosimi 39 kPa ga ortdi. Gaz dastlab qanday bosimda bo‘lgan.

Berilgan:

$T_1 = 280\ \text{K}$

$T_2 = 540\ \text{K}$

$\Delta p = 39\ \text{kPa}$

$V = \text{const}$

Topish kerak:

$p_1 - ?$

Formula:

$V = const$ da $\frac{p_1}{p_2} = \frac{T_1}{T_2}$ $\quad p_2 = p_1 + \Delta p$

$$\frac{p_1}{p_1+\Delta p} = \frac{T_1}{T_2} \Rightarrow \quad p_1 \cdot T_2 = (p_1+\Delta p) \cdot T_1$$

$$p_1 \cdot T_2 = p_1 \cdot T_1 + \Delta p \cdot T_1$$

$$P_1 T_2 - P_1 T_1 = \Delta P T_1$$

$$p_1 \cdot (T_2 - T_1) = \Delta p \cdot T_1 \Rightarrow p_1 = \frac{\Delta p \cdot T_1}{T_2 - T_1}$$

Yechish: $P_1 = \frac{39kPa \cdot 280K}{540K - 280K} = 42\text{kPa}$

Javob: $P_1 = 42\text{kPa}$

10-mashq 1-masala

Ballondagi gaz 17℃ temperaturada $1{,}45 \cdot 10^5$ Pa bosimga ega bo‘lsa, qanday temperaturada uning bosimi $2 \cdot 10^5$ Pa bo‘ladi?

Berilgan:

$t_1 = 17°C$

$p_1 = 1{,}45 \cdot 10^5$ Pa

$p_2 = 2 \cdot 10^5$ Pa

$V = const$

Topish kerak: $T_2 - ?$

Formula:

$V = const$ da $\frac{p_1}{p_2} = \frac{T_1}{T_2}$

$T_2 = \frac{p_2 \cdot T_1}{p_1}$

Yechish: $T_1 = 17°C + 273 = 290$ K

$$T_2 = \frac{2 \cdot 10^5 \text{ Pa} \cdot 290 \text{ K}}{1{,}45 \cdot 10^5 \text{Pa}} = 400 \text{ K}$$

Javob: $T_2 = 400$ K

24-test (46-bet)

Yopiq idishdagi temperaturasi $-96℃$ bo‘lgan ideal gazni $81℃$ gacha qizdirganda uning bosimi necha marta ortadi?

A) 3 **B) 2** **C) 1,18** **D) 2,21**

Berilgan:

$t_1 = -96℃$

$t_2 = 81℃$

$V = const$

Topish kerak: $\frac{p_2}{p_1} - ?$

Formula:

$V = const$ da $\frac{p_1}{p_2} = \frac{T_1}{T_2}$

$\frac{p_2}{p_1} = \frac{T_2}{T_1}$

$T = t + 273$

Yechish: $T_1 = -96℃ + 273 = 177\ K$

$T_2 = 81℃ + 273 = 354\ K$

$\frac{p_2}{p_1} = \frac{354\ K}{177\ K} = 2$

Javob: 2 marta ortadi

Masala

Ballondagi gaz temperaturasi 400 K ga teng bo‘lib, uning bosimi 3 marta ortgan bo‘lsa, gazning oxirgi temperaturasini aniqlang (K).

A) 450 **B) 1200** **C) 750** **D) 600**

Berilgan:

$T_1 = 400\ \text{K}$

$\frac{p_2}{p_1} = 3$

$V = const$

Formula:

$V = const$ da $\frac{p_1}{p_2} = \frac{T_1}{T_2}$

$T_2 = \frac{p_2 \cdot T_1}{p_1} = \frac{p_2}{p_1} \cdot T_1$

Topish kerak:

$T_2 - ?$

Yechish: $T_2 = 3 \cdot 400\ \text{K} = 1200\ \text{K}$

Javob: $T_2 = 1200\ \text{K}$

Mustaqil bajarish uchun topshiriqlar

1. Mavzuga doir savollarga javob yozish. (38-bet)

2. Agar cho‘g‘lanma lampochka yonganda uning temperaturasi 17℃ dan 360℃ gacha ko‘tarilsa, uning ichidagi gaz bosimi qanday o‘zgaradi? (10-mashq 2-masala).

3. Gaz 300 K dan 420 K gacha izoxorik qizdirilganda uning bosimi 52 kPa ga ortdi. Gaz dastlab qanday bosimda bo‘lgan? (10-mashq 2-masala).